

● MEDIUM

● ALGEBRA

Systems of two linear equations in two variables

04

10 questions · ~15 min

Q1

ALGEBRA

$$y = 3x + 9$$

$$3y = 8x - 6$$

The solution to the given system of equations is x, y . What is the value of $x - y$?

- A. -123
- B. -33
- C. 3
- D. 57

Q2

ALGEBRA

A company that provides whale-watching tours takes groups of 21 people at a time. The company's revenue is 80 dollars per adult and 60 dollars per child. If the company's revenue for one group consisting of adults and children was 1,440 dollars, how many people in the group were children?

- A. 3
- B. 9
- C. 12
- D. 18

The score on a trivia game is obtained by subtracting the number of incorrect answers from twice the number of correct answers. If a player answered 40 questions and obtained a score of 50, how many questions did the player answer correctly?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$\frac{1}{2}y = 4$$

$$x - \frac{1}{2}y = 2$$

The system of equations above has solution (x, y) . What is the value of x ?

A. 3

B. $\frac{7}{2}$

C. 4

D. 6

Q5

ALGEBRA

$$-15x + 25y = 65$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which of the following could be the second equation in the system?

- A. $12x + 20y = 52$
- B. $12x + 20y = -52$
- C. $-12x + 20y = 52$
- D. $-12x + 20y = -52$

Q6

ALGEBRA

$$y = \frac{2}{7}x + 3$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. If the second equation in the system is $y = mx + b$, where m and b are constants, what is the value of b ?

- A. -3
- B. $-\frac{1}{3}$
- C. $\frac{1}{3}$
- D. 3

$$8x + y = 5$$

$$y = 9x + 1$$

The solution to the given system of equations is x, y . What is the value of x ?

A. -6

B. $\frac{4}{17}$

C. $\frac{6}{17}$

D. 4

Q8

GRID-IN ALGEBRA

$$x + 3y = 29$$

$$3y = 11$$

The solution to the given system of equations is x, y . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

Which of the following systems of linear equations has no solution?

A. $y = 6x + 3$
 $y = 6x + 9$

B. $y = 10$
 $y = 10x + 10$

C. $y = 14x + 14$
 $y = 10x + 14$

D. $x = 3$
 $y = 10$

Q10

ALGEBRA

$$\begin{aligned}4x + 5y &= 100 \\5x + 4y &= 62\end{aligned}$$

If the system of equations above has solution (x, y) , what is the value of $x + y$?

A. 0

B. 9

C. 18

D. 38